

Did you see a driver in that car

Here are some of the autonomous cars expected to hit the roads in the very near future:

- Tesla Motors' self-steering Model S sedan
- General Motors' hands-free highway-driving Cadillac
- Mercedes-Benz hands-free system

Apart from this, numerous car makers ranging from Honda and Hyundai to Toyota are gradually introducing some of their autonomous steer-and-stop features in cars in a phased manner. Google's autonomous cars are also seen driving around Silicon Valley, California, and it is believed that these will be commercially available within five years.

heads-up display technology will add to this unobstructed view, by providing information to keep the driver's full attention on the road. For example, the movement of others on the road could be highlighted with an onscreen halo moving across the car's virtual windshield.

Further to these navigation aids, a ghost car could be projected in front of the car for the driver to follow, in case of difficulty in navigating through busy urban roads.

These concepts are part of a suite of connected technologies being developed by Jaguar Land Rover to improve road safety. According to the company's press reports, the full potential of this windshield would be delivered by connecting it to the cloud.

More power to electric vehicles

While smartness is one of the dominant quests in the automotive industry, the other is the quest for efficient electric vehicles that could reduce environmental hazards of the ever-increasing number of vehicles plying on roads today. Understandably, several research initiatives are targeted at this.

Tiny yet powerful nanopores. In a US Department of Energy-funded research at University of Maryland, the team has invented a tiny structure that includes all components of a battery, representing what they claim to be the ultimate miniaturisation of energy-storage components.

Called nanopore, this structure features a tiny hole in a ceramic sheet that holds the electrolyte to carry the electrical charge between the nanotube electrodes at either

end. The battery can be fully charged in 12 minutes and can be recharged thousands of times.

Millions of these nanopores can be combined into one larger battery, the size of a postage stamp. Since all nanopores are sized uniformly, it is possible to cram innumerable units into a single package. Such thin, small and efficient batteries are expected to revolutionise electric vehicles (EVs) as it is possible to store lots of energy within a small footprint and in a very light package, too.

Pop goes the crystal, catching all the light. Recently, a team of scientists led by Prof. Jagadeesh J. Vittal at National University of Singapore (NUS) discovered a chemical reaction that can make microscopic crystals leap distances of hundred times their own size when exposed to ultraviolet (UV) light. This distance is comparable to a human jumping several metres.

In simple terms, this is the conversion of light energy into mechanical motion. But, what makes it so exciting is that, it is the first time scientists have found such a photosensitive reaction in solids, which makes it amenable to several applications. For instance, it could result in a fresh new approach for directly converting solar power into mechanical motion, such as the movement of light-driven actuators and mechanical devices.

Another positive note in this research is that, this phenomenon comes into effect even when crystals are irradiated with weak UV light. Perhaps one day, this would lead to EVs driven directly by the sun with not many middle men in between.

Nano gives superpower to super-cars. Another nanotech breakthrough in this space comes from Queensland University of Technology (QUT). Here, researchers have developed lightweight supercapacitors that can be combined with regular batteries to give a power boost to electric cars.

An electrolyte is sandwiched between two all-carbon electrodes to make a thin and strong film with high power density. These film-like supercapacitors can be easily embedded into a car's body panels, roof, doors and so on. Being super-efficient and capable of covering a large area, these can store enough energy to charge a car's battery in just a few minutes.

According to a press report, "Supercapacitors offer a high power output in a short time, meaning, a faster acceleration rate of the car and a charging time of just a few minutes, compared to several hours for a standard electric car battery."

While currently supercapacitors are used along with li-ion batteries, in the future, they hope that supercapacitors will be capable of storing more energy than li-ion batteries and releasing this energy up to ten times faster, so that a car can be entirely powered by the supercapacitors in its panels. Expected to become a reality in a decade or so, such a car can run up to 500km on a single full charge.

Indeed, so much is happening in the automotive industry to improve driving comfort, safety and energy-efficiency that, it is mind-baffling.

In an amazing opinion piece in *The Guardian*, Tom Chatfield writes, "For those of us who do drive, the moment we get behind the wheel, we are embarking upon the most skilled, perilous and logistically fraught act of our daily lives. We are sitting inside the most expensive hunk of consumer technology we own." No wonder, he believes that smartcars will become much more popular than wearables. Yes, we are at the threshold of the age of the drivables! ●